

axis pointing against the direction of gravity. Using equation (7.5.6), we find that the pressure distribution in the liquid is given by

$$p = P_0 + \frac{1}{2} \rho \Omega^2 x + \rho g_x x. \quad (7.5.8)$$

(a) Show that, when the surface tension is insignificant, the shape of the free surface is described by the equation

$$x = \frac{\Omega^2 \sigma^2}{2g} + c, \quad (7.5.9)$$

where c is a constant, and g is the acceleration of gravity.

(b) Derive a differential equation governing the shape of the free surface in the presence of surface tension.

Computer problem

Problem c.7.5.1 *Multi-layer annular flow.*

Subroutine *tube_ann_sw_ml* in directory *04_various/uni_flow* of *FDLIB* computes the velocity profile of multi-layer swirling annular flow, as illustrated in figure 7.5.1(b). The algorithm is similar to that discussed in Section 7.1 for multi-layer channel flow.

(a) Outline and explain the numerical procedure implemented in the code.

(b) Plot the velocity profile of a three-layer flow of your choice.

7.6 Transient flow in a channel

In the preceding sections, we have studied steady unidirectional flows with rectilinear or circular streamlines. In the remainder of this chapter, we turn our attention to the corresponding unsteady flows produced by sudden or oscillatory boundary motion, tilting, or application of a pressure gradient. Our analysis will continue to be based on the assumption of unidirectional motion with rectilinear or circular streamlines, but the velocity will be allowed to change in time.

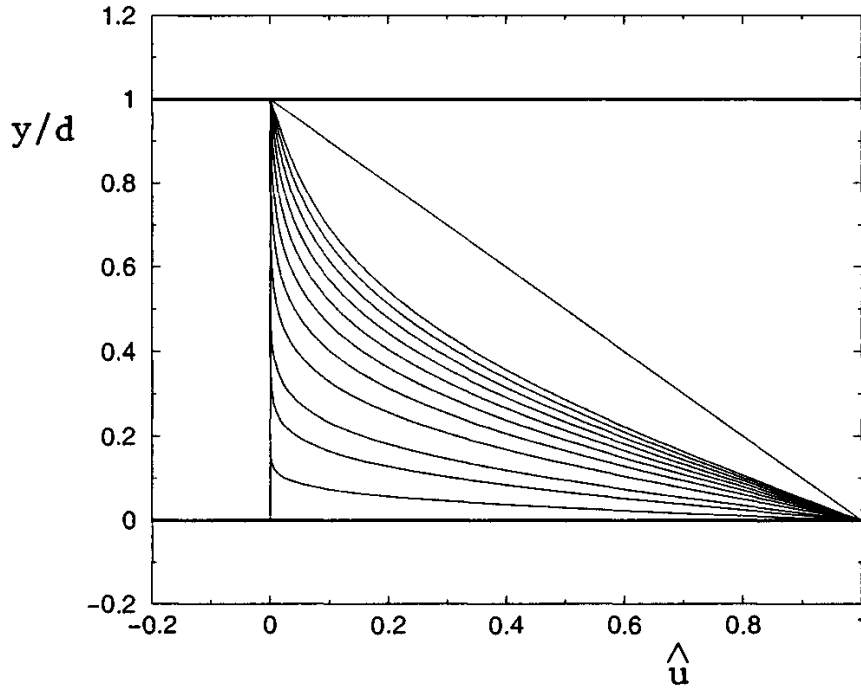


Figure 7.6.1 Transient Couette flow in a channel with parallel-sided walls due to the impulsive translation of the lower wall, at dimensionless times $\hat{t} \equiv t \nu/d^2 = 0.001, 0.005, 0.010, 0.020, 0.030, \dots, 0.14, 0.15$; the straight line is the linear profile established at long times.

7.6.1 Transient Couette flow

To begin, we consider flow in a two-dimensional channel with parallel walls separated by the distance d , where the lower wall is suddenly set in motion parallel to itself along the x axis with constant velocity U , while the upper wall is held stationary, as illustrated in figure 7.6.1

Working as in Section 7.1, we find that, in the absence of a pressure gradient along the x axis, the pressure assumes its hydrostatic distribution. The x component of the equation of motion provides us with a partial differential equation for the x component of the velocity,

$$\rho \frac{\partial u_x}{\partial t} = \mu \frac{\partial^2 u_x}{\partial y^2}, \quad (7.6.1)$$

which is to be solved subject to the initial condition

$$u_x = 0 \quad \text{for } 0 < y \leq d \text{ at } t = 0, \quad (7.6.2)$$

and to the no-slip boundary conditions

$$u_x(y = 0) = U, \quad u_x(y = d) = 0, \quad \text{for } t \geq 0. \quad (7.6.3)$$

At long times, we anticipate that the flow will reach a steady state with a linear velocity profile given by

$$u_x^{ss}(y) = U \left(1 - \frac{y}{d}\right), \quad (7.6.4)$$

as discussed in Section 7.1.

To compute the solution of (7.6.1), we consider the deviation of the transient velocity from the linear profile established at steady state, and expand it in a Fourier series with respect to y , obtaining

$$u_x(y, t) = u_x^{ss}(y) + \sum_{n=1}^{\infty} A_n(t) \sin \frac{n\pi y}{d}. \quad (7.6.5)$$

The arguments of the trigonometric functions on the right-hand side were designed to satisfy the boundary conditions (7.6.3) at all times. To compute the functions $A_n(t)$, we substitute expansion (7.6.5) into the governing equation (7.6.1), and carry out the time and space differentiation to find

$$\sum_{n=1}^{\infty} \left[\rho \frac{dA_n(t)}{dt} + \mu \left(\frac{n\pi}{d}\right)^2 A_n(t) \right] \sin \frac{n\pi y}{d} = 0. \quad (7.6.6)$$

To ensure the satisfaction of this equality for any value of y , we set the expression enclosed by the square brackets equal to zero, and thereby obtain a first-order linear differential equation for $A_n(t)$ whose solution is

$$A_n(t) = b_n \exp\left(-\frac{n^2\pi^2\nu}{d^2} t\right), \quad (7.6.7)$$

where $\nu = \mu/\rho$ is the kinematic viscosity, and b_n is a constant. Substituting expression (7.6.7) into (7.6.5), we find

$$u_x(y, t) = u_x^{ss}(y) + \sum_{n=1}^{\infty} b_n \exp\left(-\frac{n^2\pi^2\nu}{d^2} t\right) \sin \frac{n\pi y}{d}. \quad (7.6.8)$$

To compute the constants b_n , we require that (7.6.8) satisfies the initial condition (7.6.2),

$$u_x(y, t=0) = U \left(1 - \frac{y}{d}\right) + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi y}{d} = 0. \quad (7.6.9)$$

The solution follows readily by use of the trigonometric identity

$$\int_0^d \sin \frac{n\pi y}{d} \sin \frac{m\pi y}{d} dy = \begin{cases} 0 & \text{if } n = 0 \text{ or } m = 0 \\ d/2 & \text{if } n = m \neq 0 \\ 0 & \text{if } n \neq m \end{cases} \quad (7.6.10)$$

Multiplying the middle and right-hand side of (7.6.9) with $\sin(m\pi y/d)$, integrating with respect to y from 0 to d , using identities (7.6.10), and then switching m to n , we find

$$b_n = -\frac{2U}{d} \int_0^d \left(1 - \frac{y}{d}\right) \sin \frac{n\pi y}{d} dy = -\frac{2U}{n\pi}. \quad (7.6.11)$$

The velocity profile (7.6.8) takes the final form

$$u_x(y, t) = U \left[1 - \frac{y}{d} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \exp\left(-\frac{n^2 \pi^2 \nu}{d^2} t\right) \sin \frac{n\pi y}{d} \right]. \quad (7.6.12)$$

Subroutine *chan_2d_imp* in directory *04_various/uni_flow_u* of *FDLIB* evaluates the velocity given in (7.6.12). A sequence of developing profiles of the reduced velocity $\hat{u} \equiv u_x/U$ is shown in figure 7.6.1.

At long times, the summed terms on the right-hand side of (7.6.12) decay at an exponential rate, and the linear profile of the steady Couette flow expressed by the first two terms within the angular brackets on the right-hand side dominates. The elapsed time where steady flow has virtually been established, denoted by t_s , may be estimated by setting the magnitude of the argument of the exponential term on the right-hand side of (7.6.12) equal to unity for the lowest value of n , $n = 1$, obtaining

$$t_s \simeq \frac{d^2}{\pi^2 \nu}. \quad (7.6.13)$$

Apart from the factor π^2 in the denominator, this estimate could have been deduced at the outset on the basis of dimensional analysis.

Diffusion of vorticity

The strength of the vorticity in unidirectional flow is given by $\omega_z = -\partial u_x / \partial y$. Differentiating both sides of (7.6.1) with respect to y , we find that the evolution of the vorticity is governed by equation (7.6.1), with u_x replaced by ω_z on both sides.

Figure 7.6.1 illustrates that, as soon as the lower wall begins translating, a thin layer of highly rotational fluid is established near the wall. The vorticity then diffuses away from the wall to occupy the whole of the channel. When steady state has been established, the vorticity has the uniform value U/d . This process exemplifies how vorticity enters a flow across a boundary during a period of transient motion.

7.6.2 Flow due to the impulsive motion of a plate

A more detailed analysis of the transient Couette flow at short times may be carried out by neglecting the presence of the upper wall, and concentrating on the behavior of the flow near the moving wall. This is done by setting $d = \infty$ in the initial condition (7.6.2) and boundary conditions (7.6.3). cursory inspection reveals that the Fourier-series solution (7.6.12) is no longer useful in this limit.

The absence of a length scale associated with a channel of infinite width suggests that the time and space dependencies must combine into a unified variable which can be nondimensionalized by the kinematic viscosity. Recalling that the kinematic viscosity has units of length squared divided by time, we introduce the dimensionless variable

$$\eta \equiv \frac{y}{\sqrt{\nu t}}, \quad (7.6.14)$$

ranging from zero to infinity, and express the velocity in the form

$$u_x(y, t) = U f(\eta), \quad (7.6.15)$$

where $f(\eta)$ is a function of a single variable. This functional form implies that the velocity, as seen by an observer who finds herself at the position $y = \sqrt{\nu t}$, and is thus traveling upward with velocity $dy/dt = \sqrt{\nu/(4t)}$, remains constant. The initial and boundary conditions are satisfied provided that the function $f(\eta)$ satisfies the boundary condition $f(\eta = 0) = 1$, and the far-field condition $f(\eta = \infty) = 0$.

Substituting (7.6.15) into (7.6.1), we find

$$\rho \frac{\partial u_x}{\partial t} = \rho U \frac{df}{d\eta} \frac{\partial \eta}{\partial t} = \mu U \frac{\partial^2 u_x}{\partial y^2} = \mu \frac{d}{d\eta} \left(\frac{\partial \eta}{\partial y} \frac{df}{d\eta} \right) \frac{\partial \eta}{\partial y}. \quad (7.6.16)$$

Carrying out the differentiations, we obtain the second-order *nonlinear* ordinary differential equation

$$\frac{d^2 f}{d\eta^2} = -\frac{1}{2} \eta \frac{df}{d\eta}. \quad (7.6.17)$$

To compute the solution, we recast (7.6.17) into the form

$$\frac{d}{d\eta} \left(\ln \frac{df}{d\eta} \right) = -\frac{1}{2} \eta. \quad (7.6.18)$$

which may be integrated to yield

$$\frac{df}{d\eta} = A \exp\left(-\frac{\eta^2}{4}\right), \quad (7.6.19)$$

where A is a constant. Carrying out a second integration, we find

$$\begin{aligned} f(\eta) &= A \int_0^\eta \exp\left(-\frac{q^2}{4}\right) dq + B \\ &= 2A \int_0^{\eta/2} \exp(-v^2) dv + B, \end{aligned} \quad (7.6.20)$$

where B is a new integration constant, q is a dummy variable of integration, and $v \equiv q/2$.

The integral on the right-hand side of (7.6.20) is not available in analytical form. As an alternative, we express it in terms of the *error function* defined as

$$\operatorname{erf}(z) \equiv \frac{2}{\sqrt{\pi}} \int_0^z \exp(-v^2) dz, \quad (7.6.21)$$

or in terms of the *complementary error function* defined as

$$\operatorname{erfc}(z) = 1 - \operatorname{erf}(z). \quad (7.6.22)$$

Graphs of these functions are shown in figure 7.6.2. As z tends to infinity, the error function tends to the asymptotic value of unity; correspondingly, the complementary error function tends to vanish.

The frequent occurrence of the error function in various branches of mathematical physics has motivated its tabulation and representation in terms of various approximations. Subroutine *error-f* in directory *01_num_meth/12_spec_fnc* of *FDLIB* evaluates the error function and the complementary error function using an accurate algebraic approximation.

Returning to (7.6.20), we express the right-hand side in terms of the error function, and require $f(\eta = 0) = 1$, and $f(\eta = \infty) = 0$ to obtain $A = -1/\sqrt{\pi}$ and $B = 1$, yielding the velocity profile

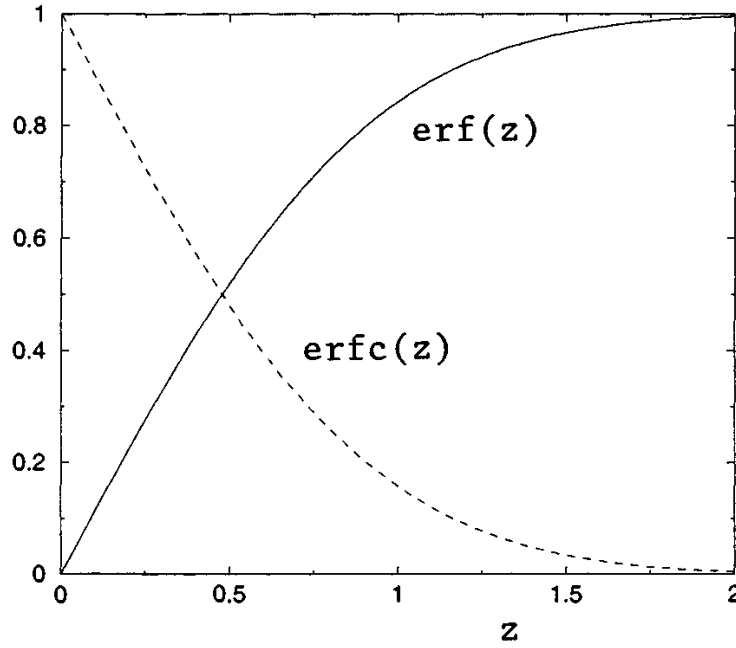


Figure 7.6.2 Graphs of the error function (solid line) and complementary error function (dashed line) defined in equations (7.6.20) and (7.6.21), computed using algebraic approximations.

$$u_x(y, t) = U f(\eta) = U \operatorname{erfc}\left(\frac{\eta}{2}\right). \quad (7.6.23)$$

The accompanying vorticity profile is given by

$$\omega_z(y, t) = -\frac{\partial u_x}{\partial y} = -U \frac{df}{d\eta} \frac{\partial \eta}{\partial y} = \frac{U}{\sqrt{\pi \nu t}} \exp\left(-\frac{y^2}{4\nu t}\right). \quad (7.6.24)$$

This expression illustrates explicitly the singular nature of the vorticity at the initial instant, and the spreading of the vorticity away from the moving wall due to viscous diffusion at later times.

The wall shear stress is given by

$$\sigma_{xy}(y = 0, t) = \mu \left(\frac{\partial u_x}{\partial y}\right)_{y=0} = -\frac{\mu U}{\sqrt{\pi \nu t}}. \quad (7.6.25)$$

This expression illustrates that an unphysical singularity occurs at the initial instant, as soon as the plate starts moving. In reality, a plate cannot start moving with constant velocity in an impulsive fashion; instead, its velocity must increase gradually from zero to the final value over a non-infinitesimal period of time, however small.

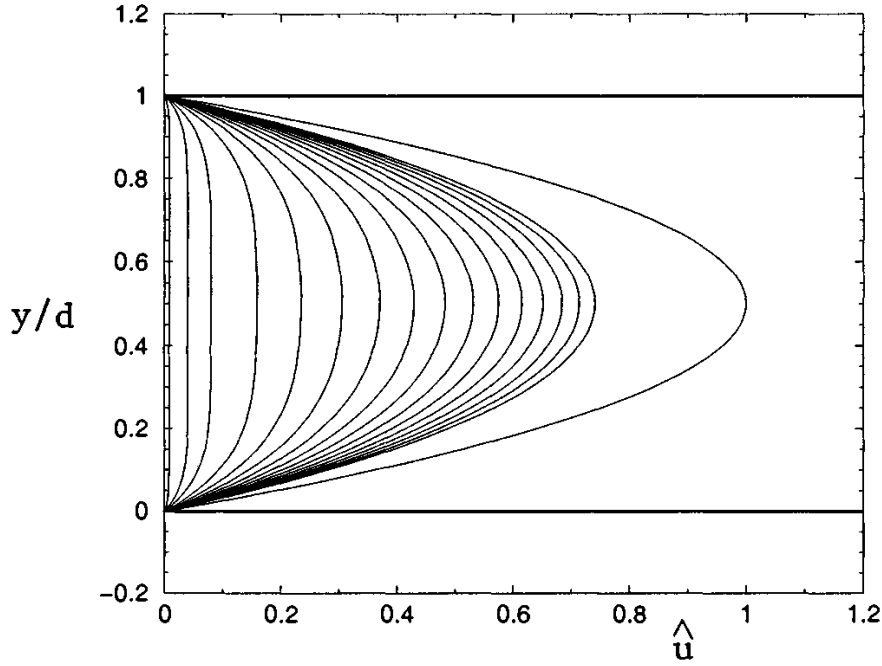


Figure 7.6.3 Transient pressure- or gravity-driven flow in a channel with parallel-sided walls separated by the distance d , due to the sudden application of a constant pressure gradient or sudden tilting, at dimensionless times $\hat{t} \equiv t \nu / d^2 = 0.001, 0.005, 0.010, 0.020, 0.030, \dots$. The parabolic line corresponds to the Hagen flow established at long times.

7.6.3 Transient pressure- or gravity-driven flow

As a second case study, we consider flow in a channel with stationary walls, where the motion is due to the sudden application of a constant pressure gradient or sudden tilting of the channel walls by a certain angle.

The x component of the equation of motion provides us with a partial differential equation for the x component of the velocity,

$$\rho \frac{\partial u_x}{\partial t} = \chi + \mu \frac{\partial^2 u_x}{\partial y^2} + \rho g_x. \quad (7.6.26)$$

The pressure field is given by equation (7.1.3), where the pressure gradient χ is a constant.

Working as in the case of transient Couette flow discussed previously in this section, and requiring that the velocity at the lower and upper wall vanishes at all times, we derive the transient velocity profile

$$u_x(y, t) = \frac{\chi + \rho g_x}{2\mu} [y(d - y) - \frac{8d^2}{\pi^3} \sum_{n=1,3,\dots}^{\infty} \frac{1}{n^3} \exp(-\frac{n^2\pi^2\nu}{d^2} t) \sin \frac{n\pi y}{d}]. \quad (7.6.27)$$

Subroutine *chan_2d_trans* in subdirectory *04_various/uni_flow_u* of *FDLIB* evaluates the profile at a specified time. A sequence of developing profiles of the dimensionless velocity $\hat{u} \equiv u_x/U$ is shown in figure 7.6.3, where U is the maximum velocity of the steady Hagen flow established at long times.

Problems

Problem 7.6.1 *Transient pressure-driven flow.*

Derive the transient velocity profile (7.6.27).

Problem 7.6.2 *Flow due to a constant shear stress.*

Show that the velocity field due to the application of a constant shear stress of magnitude τ on the planar boundary of a semi-infinite fluid occupying the upper half-space $y \geq 0$, is given by

$$u(y, t) = \frac{\tau}{\mu} \sqrt{\nu t} \left[\frac{2}{\sqrt{\pi}} \exp(-\frac{\eta^2}{4}) - \eta \operatorname{erfc}(\frac{\eta}{2}) \right], \quad (7.6.28)$$

where $\eta \equiv y/\sqrt{\nu t}$. Discuss the asymptotic behavior at long times.

7.7 Oscillatory flow in a channel

In Section 7.6, we discussed transient Couette flow and transient pressure- or gravity-driven flow. In all cases, vorticity enters the flow through the boundaries, and then diffuses to occupy the whole of the domain of flow. In this section, we consider the corresponding problems for oscillatory flow. The important new feature is that vorticity of alternating sign enters the flow across the boundaries, and cancellation by diffusion prevents the establishment of motion far from the boundaries.

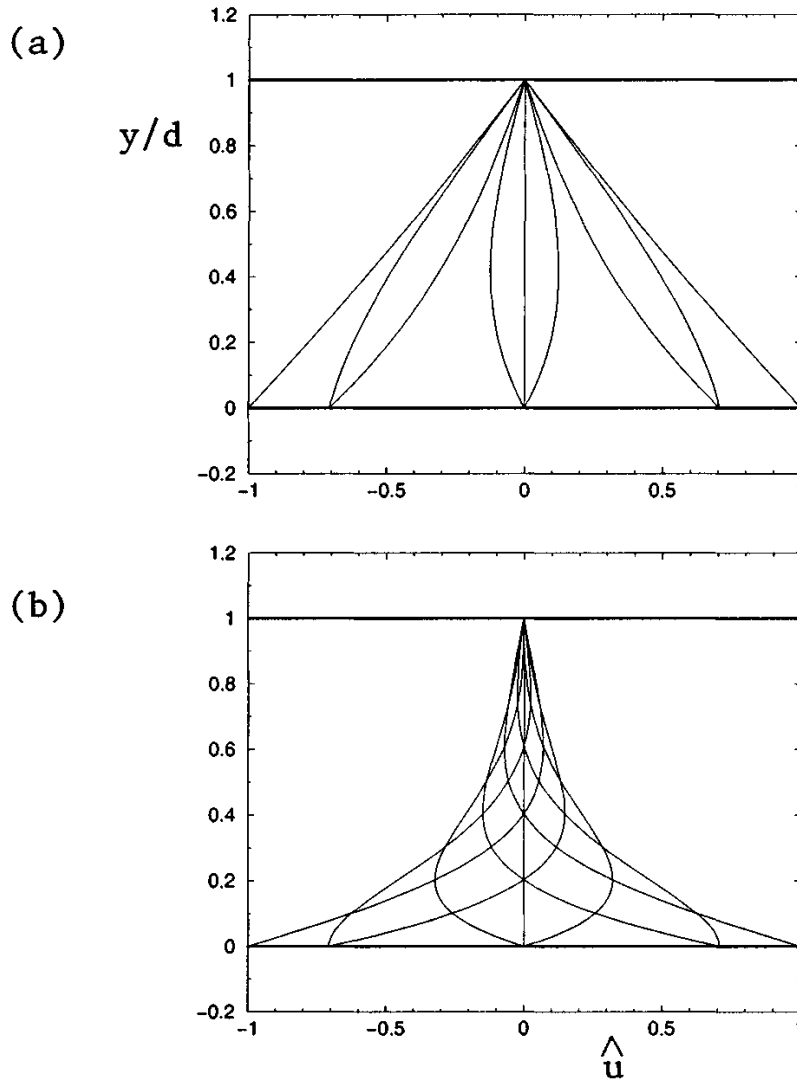


Figure 7.7.1 Velocity profiles of oscillatory Couette flow in a two-dimensional channel, at phase angles $\Omega t = 0, 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 1.75$, and 2.0 , for Womersley number (a) $N_W = \sqrt{2}$, and (b) $\sqrt{30}$.

7.7.1 Oscillatory Couette flow

Suppose that the lower wall of a two-dimensional channel oscillates parallel to itself with velocity $U \cos(\Omega t)$, while the upper wall is stationary, thereby generating a time-periodic unidirectional flow, as illustrated in figure 7.7.1. The flow is governed by the unsteady diffusion equation (7.6.1), subject to the boundary conditions

$$u_x(y = 0) = U \cos(\Omega t), \quad u_x(y = d) = 0, \quad (7.7.1)$$

for $t \geq 0$.

Anticipating that the flow will be periodic in time, we express the velocity in the form

$$u_x(y, t) = U [f_c(y) \cos(\Omega t) + f_s(y) \sin(\Omega t)]. \quad (7.7.2)$$

To compute the functions $f_c(y)$ and $f_s(y)$, we substitute (7.7.2) into (7.6.1), carry out the time differentiation, collect the coefficients of the sines and cosines, and then set the compiled expressions equal to zero to obtain two linear coupled ordinary differential equations,

$$\frac{d^2 f_s}{dy^2} = -\frac{\Omega}{\nu} f_c, \quad \frac{d^2 f_c}{dy^2} = \frac{\Omega}{\nu} f_s, \quad (7.7.3)$$

where $\nu \equiv \mu/\rho$ is the kinematic viscosity.

To expedite the solution, we introduce the imaginary unit i , defined such that $i^2 = -1$, and the complex function

$$f(y) \equiv f_c(y) + i f_s(y). \quad (7.7.4)$$

Subject to this definition, the two equations in (7.7.3) combine into the single complex form

$$\frac{d^2 f}{dy^2} = -i \frac{\Omega}{\nu} f. \quad (7.7.5)$$

The real and imaginary parts of the complex equation (7.7.5) are equal, respectively, to the second and first of equations (7.7.3). Moreover, using the Euler decomposition $\exp(-i \Omega t) = \cos(\Omega t) - i \sin(\Omega t)$, we find that u_x is given by the real part of the complex function

$$w(y, t) \equiv U f(y) \exp(-i \Omega t). \quad (7.7.6)$$

The imaginary part of $w(y, t)$ also satisfies the governing equation (7.6.1).

The solution of (7.7.5) is readily found to be

$$f(y) = A \exp(y \sqrt{-i \frac{\Omega}{\nu}}) + B \exp(-y \sqrt{-i \frac{\Omega}{\nu}}), \quad (7.7.7)$$

where A and B are two complex constants, with the understanding that, if β is a positive or negative real number, then

$$\begin{aligned}
\exp(\beta\sqrt{-i}) &= \exp\left[\frac{\beta}{\sqrt{2}}(-1+i)\right] = \exp\left(-\frac{\beta}{\sqrt{2}}\right) \exp\left(i\frac{\beta}{\sqrt{2}}\right) \\
&= \exp\left(-\frac{\beta}{\sqrt{2}}\right) \left[\cos\left(\frac{\beta}{\sqrt{2}}\right) + i \sin\left(\frac{\beta}{\sqrt{2}}\right)\right].
\end{aligned} \tag{7.7.8}$$

The no-slip boundary conditions (7.7.1) require $f(0) = 1$ and $f(d) = 0$. Using these equations to evaluate A and B , we find

$$\begin{aligned}
A &= \frac{\exp(-d\sqrt{-i\frac{\Omega}{\nu}})}{\exp(-d\sqrt{-i\frac{\Omega}{\nu}}) - \exp(d\sqrt{-i\frac{\Omega}{\nu}})} = \frac{1}{1 - \exp(2d\sqrt{-i\frac{\Omega}{\nu}})}, \\
B &= 1 - A.
\end{aligned} \tag{7.7.9}$$

Substituting these expressions into (7.7.6), and rearranging, we derive the complex function

$$f(y) = \frac{\exp(y\sqrt{-i\frac{\Omega}{\nu}}) - \exp[(2d-y)\sqrt{-i\frac{\Omega}{\nu}}]}{1 - \exp(2d\sqrt{-i\frac{\Omega}{\nu}})}, \tag{7.7.10}$$

whose real and imaginary parts provide us with the velocity according to equations (7.7.4) and (7.7.2). Subroutine *chan_2d_osc* in directory *04_various/uni_flow_u* of *FDLIB* evaluates the velocity profile for a specified angular frequency and reduced temporal phase angle Ωt .

The argument of the exponentials on the right-hand side of (7.7.10) suggests that the structure of the flow is determined by the reduced angular frequency expressed by the *Womersley number*

$$N_W \equiv d\sqrt{\frac{\Omega}{\nu}}. \tag{7.7.11}$$

Figure 7.7.1 illustrates profiles of the dimensionless velocity $\hat{u} \equiv u_x/U$ at $\Omega t = 0, 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 1.75$, and 2.0 , for $N_W = \sqrt{2}$ and $\sqrt{30}$. When the Womersley number is small, the flow evolves in quasi-steady fashion, and the velocity profile is nearly linear at all times. As the Womersley number is raised, the motion is confined within a boundary layer close to the moving wall, and the rest of the fluid is virtually stationary.

7.7.2 Rayleigh's oscillating plate

To investigate the behavior at high frequencies or large channel widths, we consider the structure of the flow at large values of the Womersley number. In this limit, the second term in the numerator and the second term in the denominator on the right-hand side of (7.7.10) become exponentially small, leaving the simplified form

$$\begin{aligned} f(y) &= \exp\left(y \sqrt{-i \frac{\Omega}{\nu}}\right) \\ &= \exp\left(-y \sqrt{\frac{\Omega}{2\nu}}\right) \left[\cos\left(y \sqrt{\frac{\Omega}{2\nu}}\right) + i \sin\left(y \sqrt{\frac{\Omega}{2\nu}}\right)\right]. \end{aligned} \quad (7.7.12)$$

The associated velocity profile is given by

$$\begin{aligned} u_x(y, t) &= U \exp\left(-y \sqrt{\frac{\Omega}{2\nu}}\right) \\ &\quad \times \left[\cos(\Omega t) \cos\left(y \sqrt{\frac{\Omega}{2\nu}}\right) + \sin(\Omega t) \sin\left(y \sqrt{\frac{\Omega}{2\nu}}\right)\right] \\ &= U \exp\left(-y \sqrt{\frac{\Omega}{2\nu}}\right) \cos\left(\Omega t - y \sqrt{\frac{\Omega}{2\nu}}\right). \end{aligned} \quad (7.7.13)$$

Subroutine *plate_osc* in directory *04_various/uni_flow_u* of *FDLIB* evaluates this velocity profile at a specified time.

Expression (7.7.13) shows that the magnitude of the velocity decays at an exponential rate with respect to distance from the oscillating wall. The region where substantial fluid motion is established is identified as the *Stokes boundary layer*. A rough measure of the thickness of this boundary layer, denoted by δ , is the distance where the magnitude of the argument of the exponential on the right-hand side of (7.7.13) is equal to unity, yielding

$$\delta = \sqrt{\frac{2\nu}{\Omega}}. \quad (7.7.14)$$

Aside from the factor 2 in the numerator, this estimate could have been deduced at the outset on the basis of dimensional analysis.

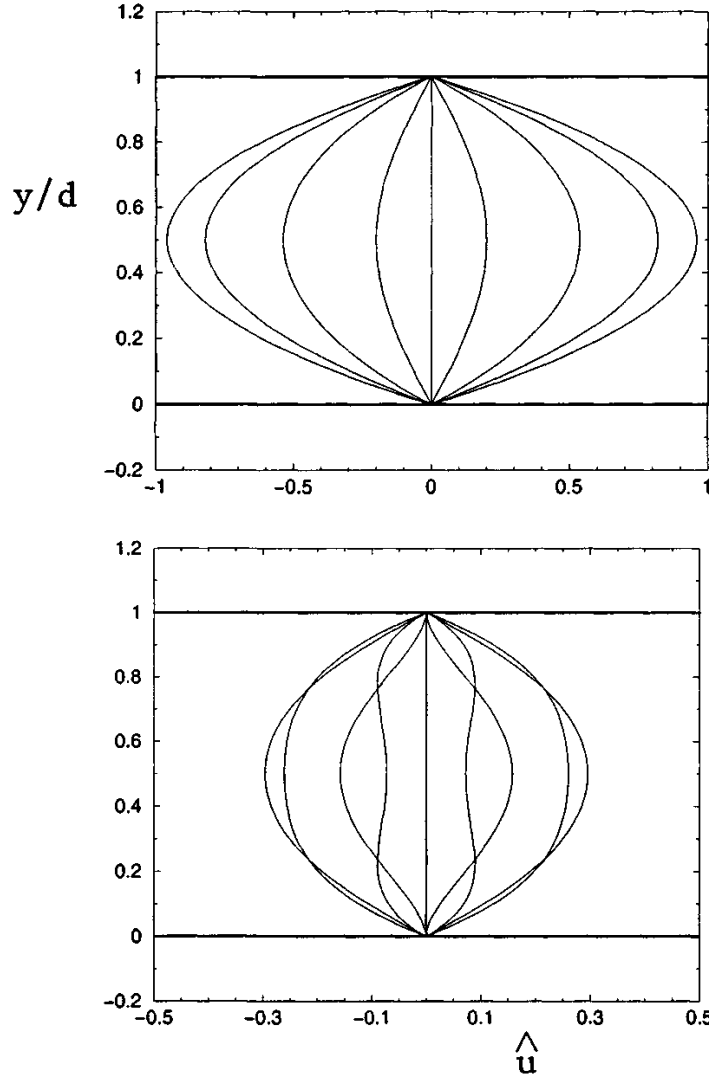


Figure 7.7.2 Velocity profiles of pulsating pressure-driven flow in a two-dimensional channel, at phase angles $\Omega t = 0, 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 1.75, 2.0$, for Womersley number (a) $N_W = \sqrt{2}$, and (b) $\sqrt{30}$.

7.7.3 Pulsating pressure-driven flow

As a last case study, we consider pulsating flow in a channel due to an oscillatory pressure gradient. The pressure field is given by equation (7.1.3), where $\chi = \bar{\chi} \sin(\Omega t)$. Thus, the pressure gradient varies harmonically in time with angular frequency Ω and amplitude $\bar{\chi}$.

Working as previously for oscillatory Couette flow, we express the velocity in the form

$$u_x(y, t) = -\frac{\bar{\chi}}{\rho \Omega} [f_c(y) \cos(\Omega t) + f_s(y) \sin(\Omega t)], \quad (7.7.15)$$